On the Recoverability of Coalescence Phase and Polarization Angle from Gravitational-Wave Strain: A Systematically Verified Null Result

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I. Introduction

Since the first direct detection of gravitational waves from a binary black hole merger [1], deep-learning approaches to gravitational-wave parameter estimation have matured rapidly, from early proof-of-concept classifiers [2] to amortized neural posterior estimators that approach the accuracy of stochastic-sampling pipelines [3, 4] at a fraction of their latency. Most of this progress has concentrated on parameters that imprint strongly on the observed strain, i.e., the chirp mass [5], the merger time and signal-to-noise ratio [6], and the sky position [7, 8].

Two extrinsic parameters have received far less attention as direct regression targets: the orbital phase at coalescence, φ_c, and the polarization angle, ψ. There is a physical reason for this neglect. For the dominant quadrupole ($\ell = 2$, $|m| = 2$) mode of a quasi-circular compact binary, the two polarization states are [9]

$$h_+ \propto (1 + \cos^2 \iota) \cos(2\varphi(t) + 2\varphi_c),$$
$$h_\times \propto 2 \cos \iota \sin(2\varphi(t) + 2\varphi_c),$$

and the detector response mixes them through antenna patterns that depend on 2ψ [5, 9]. Both φ_c and ψ enter the strain doubled, so the analytically degenerate combination in the face-on limit ($\cos \iota \to \pm 1$) is $2\varphi_c \pm 2\psi$, with the sign set by the sign of $\cos \iota$. In the face-on limit the signal becomes circularly polarized and depends on φ_c and ψ only through this single, corrected combination, so the individual angles are exactly unidentifiable there, and only partially disentangled as the inclination moves toward edge-on. Bayesian samplers absorb this structure into broad, multimodal joint posteriors [10–13]; a point-estimate regression network has no such refuge.

This paper asks the question:

> *Can φ_c and ψ be recovered from strain alone by a deep neural network, individually, through their sum/difference combinations, or is the degeneracy effectively exact for a realistic detection-scale population?*

The answer, is a **null result**: across the five trunk architectures, two head parameterizations, and three values of a magnitude-regularization coefficient. No model could perform measurably better than random guessing on either angle, while the same networks simultaneously recovered chirp mass, merger time, signal-to-noise ratio, and in some cases, sky position with high fidelity from the same shared features.

A null result of this kind is only as strong as the effort spent trying to break it. The scientific contribution of this paper is therefore threefold. First, the null itself, tested against an explicit list of confounds: data-pipeline faults, loss-wiring faults, architecture-specific artifacts, statistical flukes, and population heterogeneity. Second, a methodological account of *how* the null was established when the standard interpretability gate never cleared — because unlike a gate failing for one model out of several, here it failed for every model at every regularization strength tried, forcing an explicit

argument for why a broader, ungated battery of evidence should be trusted in its place, a response, at engineering-loop scale, to the same replication crisis that motivates pre-registration in quantitative science more broadly [14, 15]. Third, a methodological lesson demonstrated directly within this study: a checkpoint that cleared every numeric threshold of the pre-registered interpretability gate turned out, on scatter-plot verification, to be a literal two-point mode collapse — a passing aggregate metric that is not, by itself, proof of the property it is meant to certify.

The research paper is organized as follows; Section II formalizes the problem and the circular-regression framework. Section III presents the analytic prerequisite study, rerun under the corrected antenna-pattern model. Section IV details the datasets, architectures, losses, and training protocol, including the corrected combination construction and two training-stability fixes established before the main sweep (a linear periodic-head activation, an explicit magnitude penalty). Section V verifies, on trained checkpoints, that this machinery is healthy before trusting what it produces. Section VI presents the defended null result, its verification battery, and findings. Section VII describes the pre-registered magnitude-penalty retune and its outcome. Section VIII discusses interpretation, limitations, and future work; Section IX concludes.

II. Problem Formulation

A. Targets and the degeneracy hypothesis

We consider a ten-parameter description of a compact binary coalescence signal injected into two-detector noise: chirp mass $\mathcal{M}$, mass ratio q, inclination ι, coalescence phase φ_c, polarization angle ψ, declination, right ascension, injection time, merger time within the analysis window, and network signal-to-noise ratio. Seven of these are used as supervised regression targets (mass ratio and injection time are excluded; §IV-A). The scientific targets are $\varphi_c \in [0, 2\pi)$, period 2π, and $\psi \in [0, \pi)$, period π. Inclination, chirp mass, merger time, SNR, and sky position are retained as *control heads*: parameters trained through the identical pipeline whose success or failure calibrates what the network can extract from strain when information is present.

The degeneracy hypothesis under test is: *φ_c and ψ carry no strain-only recoverable signal for this population, in this architecture and loss family*. A point-estimating network can do no better than the optimal constant prediction. For most of the population the likelihood constrains only the combination $2\varphi_c \pm 2\psi$, and the remaining direction is unconstrained. The alternative hypothesis motivating the experimental design is that even if φ_c and ψ are individually unrecoverable, their sum and difference combinations may be well-conditioned. Face-on systems constrain $2\varphi_c \pm 2\psi$ (sign by $\cos\iota$), so a network parameterized in combination space, with a curriculum that weights each combination according to how well the current inclination constrains it — quantified by the condition number of the local $(\varphi_c, \psi) \to (R, \delta)$ Jacobian, low near edge-on and diverging at face-on — might learn the recoverable structure that a naive per-angle parameterization misses.

B. Circular regression framework

Periodic targets are regressed on the unit circle [16]. An angle θ with period T is encoded as the two-vector $\mathbf{u}(\theta) = (\sin(2\pi\theta/T), \cos(2\pi\theta/T))$, so the ψ head, with $T = \pi$, physically encodes 2ψ. Each periodic head outputs a raw two-vector $\mathbf{v} \in \mathbb{R}^2$, projected onto the unit circle by a normalization layer, $\hat{\mathbf{u}} = \mathbf{v}/\sqrt{\max(\|\mathbf{v}\|^2, \varepsilon)}$, $\varepsilon = 10^{-8}$ — a clamp on the squared norm, for a stable gradient at $\mathbf{v} = \mathbf{0}$, rather than a floor on $\|\mathbf{v}\|$ itself — and trained with the cosine (circular) loss $L_{\text{circ}} = 1 - \hat{\mathbf{u}}_{\text{pred}} \cdot \mathbf{u}_{\text{true}} = 1 - \cos\Delta\theta$.

For uniformly distributed targets and any fixed prediction, the expected circular loss is exactly 1; the expected wrap-aware angular mean absolute error (MAE_{ang}) is $T/4$, i.e., $\pi/2 \approx 1.5708$ rad for φ_c and ι, $\pi/4 \approx 0.7854$ rad for ψ. These values are the null baselines against which every result in this paper is judged.

The curriculum weight used at training time is nonetheless simpler than the measured condition number alone would justify. The empirical Jacobian

sweep (§III) yields a non-monotonic, intermediate-peak curve — rising from exact zero at face-on to a maximum away from either pole, then falling back toward edge-on — and that edge-on decline cannot be distinguished from an artifact of the specific finite-difference/sky-sampling procedure used to measure it, as opposed to genuine physics. Training therefore uses the analytically motivated $w(\iota) = \sin^2 \iota$ instead: monotonic to edge-on, and committed only to the one fact that is certain, exact degeneracy at face-on, rather than encoding an unconfirmed dip in constraining power directly into the training signal.

Two further diagnostic statistics recur throughout: the circular resultant $\text{circ}_r = \|\text{mean}(\hat{\mathbf{u}})\|$, which measures mode collapse ($\text{circ}_r \to 1$ means all predictions concentrate at one angle), and $\sigma_{\text{head, target}}$, the ratio of the standard deviation of a head's raw outputs to that of the encoded targets, a monitor of raw-magnitude health with a healthy band of $[0.5, 2.0]$ — called $\text{STD}_{\text{ratio}}$ in the diagnostic chapters from §VII onward. Non-periodic heads use a Huber loss on transformed targets, and the sky-position head uses a closed-form von Mises–Fisher negative log-likelihood on the unit sphere [17]. One head cuts across this taxonomy: the inclination head keeps the periodic two-vector encoding but its loss is Huber on the raw output vector, with no normalization and no circular loss anywhere in its path. All heads share a single trunk and are combined by learned homoscedastic uncertainty weighting [18].

III. Prerequisite Study

Before any training, the combination-space hypothesis was tested analytically, using a toy detector-response model in which $h_+/h_\times$ add $2\varphi_c$ to the orbital-phase term 2Φ. A required self-check — sweeping ψ at fixed $2\varphi_c + 2\psi$, confirming (R, δ) stays exactly constant at $\iota = 0$ — passed to machine precision ($\max \text{std}(R) = 1.16 \times 10^{-14}$, $\max \text{std}(\delta) = 5.87 \times 10^{-16}$), gating trust in everything that follows.

The 100-point inclination sweep (50 points per sign of $\cos\iota$, 200 sky positions, 5,000-draw bootstrap) confirms the expected combination structure:

- for $\cos\iota > 0$, combo_B $(2\varphi_c - 2\psi)$ is the better-constrained combination throughout; for $\cos\iota < 0$, combo_A $(2\varphi_c + 2\psi)$ wins — a clean sign flip driven by the sign of $\cos\iota$;
- the preference ratio is strongest face-on, $\approx$ 1.62–1.65×, decaying monotonically to $\approx$ 1.04–1.07× at edge-on, never crossing 1.0;
- population-bootstrapped mean ratios are 1.17× (95% CI $[1.11, 1.24]$) for $\cos\iota > 0$ and 1.19× (95% CI $[1.12, 1.26]$) for $\cos\iota < 0$, with the simple two-point reference check clearing significance directly.

The sign flip described above and the monotonic decay in the preference ratio share one mechanical cause. The toy model's two antenna-weighted amplitude coefficients respond asymmetrically to inclination: $k_1(\iota) = (1 + \cos^2\iota)/2$ for h_+ stays strictly positive across $[0, \pi]$, while $k_2(\iota) = \cos\iota$ for $h_\times$ crosses zero at $\iota = \pi/2$ and changes sign on either side. Since the detector strain is $F_+h_+ + F_\times h_\times$, that sign change in $h_\times$'s own amplitude is what flips which of combo_A/combo_B the strain constrains more tightly, and the same vanishing of k_2 at edge-on is why the preference ratio closes toward 1.0× there: with $h_\times$'s contribution suppressed, the two combinations become comparably constrained rather than either one standing out. The ratio peaks at face-on instead, where k_1 and k_2 are both near their maximum magnitude and the signal is closest to circularly polarized.

The curriculum weight derivation gives $w(\text{face-on}) = 0.0000$ and $w(\text{edge-on}) = 0.1416$, with an intermediate-peak shape; training uses the analytically motivated fallback $w(\iota) = 1 - \cos^2\iota = \sin^2\iota$ rather than the fitted empirical curve. The fitted curve's edge-on value is not required by the physics above, which only guarantees the map is well short of singular there, not that its condition number should sit near this particular figure — so 0.1416 is as plausibly an artifact of the fixed-reference-point finite-difference procedure used to measure it as it is a genuine physical feature. Encoding that unconfirmed dip directly into the training signal would risk under-weighting exactly the edge-on regime where the degeneracy above is weakest;

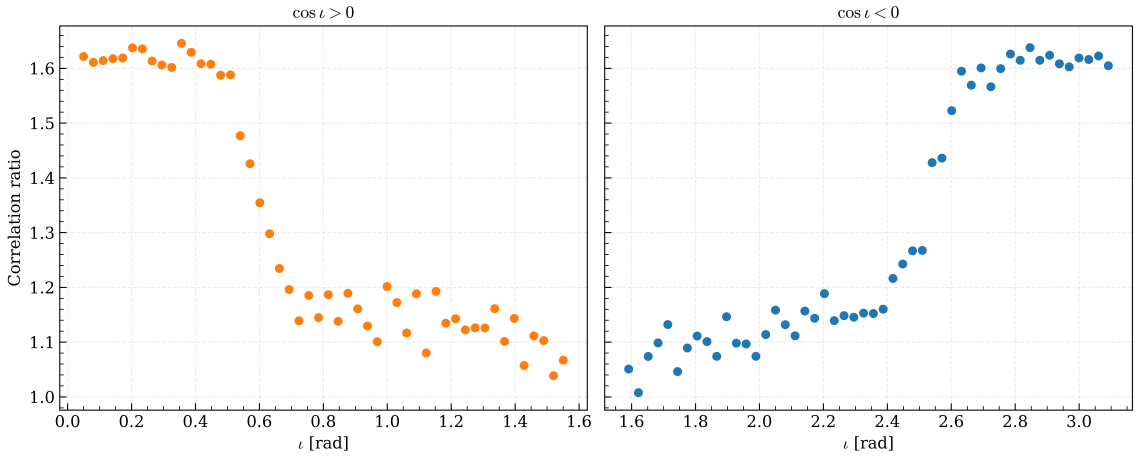


Fig. 1. Analytic check of the corrected $2\varphi_c$–2ψ degeneracy structure: ratio of the strain correlation with the well-constrained combination to the poorly-constrained one, as a function of inclination.

the fallback instead commits only to the one fact that is exact — zero weight at face-on, where the degeneracy is total — and rises monotonically from there.

IV. Methods

A. Dataset

All experiments use a single pre-generated HDF5 dataset of simulated two-detector observations: 25 000 training and 5000 validation samples, each a pair of 4096 sample noisy strain series for the H1 and L1 detectors — a 2 s window at 2048 Hz — stacked as a $(4096, 2)$ input tensor. All injections use the dominant quadrupole ($\ell = 2$, $|m| = 2$) mode only (IMRPhenomD), with no higher-order modes. Table I gives the marginal parameter distributions.

Component masses are drawn via two independent $\mathrm{Uniform}(10, 50)\, M_\odot$ values, sorted so $m_1 \geq m_2$. These drawn values were than used to derive both mass ratio and chirp mass.

The important parameter here is the inclination angle (ι). The parameter distribution is uniform in itself, over-representing face-on systems and under-representing edge-on systems relative to a truly isotropic population. Because face-on is where the φ_c/ψ degeneracy is strongest, this skew could in principle bias a *pooled* statistic — one averaged across the whole population, such as §VI's headline circular-loss numbers — mildly toward looking like a null, simply by weighting more hard-to-recover cases into the average. That possibility cannot explain the null result itself: the inclination-band stratification of §VI-G measures recovery within each band separately, so the population mix has no opportunity to act on it, and even the edge-on band, checked in isolation, clears no recovery above the materiality floor either. The strain is than fed to the model already whitened; targets are transformed per

TABLE I
Marginal parameter distributions for the injected population.

Parameter	Support	Distribution
m_1	[10, 50] $M_\odot$	Uniform pair
m_2	[10, 50] $M_\odot$	Uniform pair
$q = m_2/m_1$ $m_1 \geq m_2$	[0.20, 1.00)	Derived
$\mathcal{M}$	[8.84, 43.45] $M_\odot$	Derived
SNR	$[7, 15]$	Uniform
φ_c	$[0, 2\pi)$	Uniform
ψ	$[0, \pi)$	Uniform
ι	$[0, 2\pi)$	Uniform in ι
Right ascension	$[0, 2\pi)$	Uniform
Declination	$[-\pi/2, \pi/2]$	cos weighted

§II-B on training-split statistics.

B. Trunk architectures

Five one-dimensional trunk architectures were evaluated, all mapping the $(4096, 2)$ input to a pooled feature vector shared by every head: a temporal convolutional network (TCN) with dilated residual blocks and receptive field [19], a plain hierarchical CNN ($\text{CNN}_{\text{baseline}}$), a strided Convolution-Transformer network with learned attention pooling ($\text{CNN}_{\text{attention}}$ [20]), a multi-scale-kernel network (InceptionTime [21]), and a residual network (ResNet1D [22]).

These five were chosen as five distinct inductive-bias families, so that each represents a different hypothesis about how phase information might be encoded in the strain. Each head is a two-layer MLP (64 hidden units, ReLU) on the shared features, with a linear 2-unit output for periodic heads. Periodic heads use this linear activation rather than the more common bounded (tanh) one: tanh saturates at random initialization on this architecture, and the unit-circle normalization layer (§II-B) already projects the raw output onto the circle, so a bounded activation is unnecessary and actively harmful here.

C. Head parametrization

The baseline mode uses an independent circular-loss heads directly on φ_c and ψ, each with its own uncertainty weight. The named configuration proof-of-concept A (POC_A) is baseline mode on the TCN trunk. POC_B (also on TCN) constructs the two combination heads explicitly rather than treating φ_c and ψ as independent outputs. The φ_c head's unit vector, $\mathbf{z}_{\varphi_c}$, is first passed through a `double_angle` operation, $\mathbf{z}_{2\varphi_c} = \mathbf{z}_{\varphi_c} \otimes \mathbf{z}_{\varphi_c}$ which turns a unit vector at angle φ_c into one at angle $2\varphi_c$; the ψ head's own encoding already stores 2ψ internally (§II-B). The two combination vectors are $\mathbf{z}_A = \mathbf{z}_{2\varphi_c} \otimes \mathbf{z}_\psi$ (angle $2\varphi_c + 2\psi$) and $\mathbf{z}_B = \mathbf{z}_{2\varphi_c} \otimes \overline{\mathbf{z}_\psi}$ (angle $2\varphi_c - 2\psi$). The isotropic circular loss, sign-dependent per-sample curriculum weighting, and uncertainty-weighting scheme are fixed pipeline components, applied identically to every head. The well-constrained combination assignment (combo_B for $\cos\iota \geq 0$) is derived from this study's own prerequisite analysis (§III).

D. Magnitude penalty

The isotropic circular loss (§II-B) provides no restoring force on a periodic head's raw output magnitude $\|\mathbf{v}\|$: as $\|\mathbf{v}\|$ drifts during training, the unit-circle normalization layer's backward pass (division by $\|\mathbf{v}\|$) either crushes or explodes the angular gradient. All configurations therefore add an explicit radial regularizer on the raw, pre-normalization periodic outputs, $L_{\text{mag}} = \lambda \cdot \sum \text{mean}[(\|\mathbf{v}\| - 1)^2]$, present in every configuration from the first training run. $\lambda = 0.01$ for the main sweep (§VI), retuned to 0.05 and 0.10 in §VII.

E. Training protocol

Adam, initial learning rate 10^{-3}, `ReduceLROnPlateau` (factor 0.5, patience 5), batch size 128, 80 epochs, seed 42, uncertainty weighting with log-variance clamp ± 3, Huber $\delta = 1$. Periodic epoch-10 checkpoints were saved throughout, not only best/final weights.

F. Evaluation

Point predictions are scored by wrap-aware angular MAE, circ_r, and, for scalar heads, MAE/R^2. Statistical significance is assessed by a label-permutation bootstrap ($N = 10{,}000$ shuffles), and population heterogeneity is probed by SNR-tercile and inclination-band stratification. Two additional checks were added in response to a genuine new finding for one architecture (§VI-H) rather than planned in advance: a formal shuffle-null bootstrap and SNR-tercile stratification for the sky-position head's own angular separation.

V. Verifying the Training Machinery

A. Verification on trained checkpoints

Confirming, on the Round-1 ($\lambda = 0.01$) checkpoints, that the machinery above is healthy is a prerequisite to trusting anything it produces. True labels are clean (circular resultant ≤ 0.013

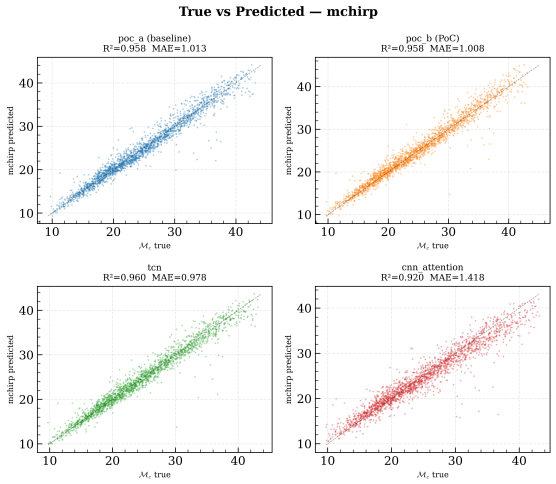


Fig. 2. Positive-control recovery (chirp mass) for the four down-selected models.

for $\varphi_c/\psi/\iota$/right ascension). Gradients reach the periodic-head weights end to end through the corrected construction: $dL/d(\text{combo}_\text{A}) = 0.4245$, dropping through the `double_angle` stage ($dL/d(\mathbf{z}_{2\varphi_c,\text{norm}}) = 0.375$) to $dL/d(\mathbf{z}_{\varphi_c,\text{raw}}) = 0.570$ and $dL/d(\mathbf{z}_{\psi,\text{raw}}) = 0.514$ — comparable in order of magnitude to the healthy inclination-head baseline, 0.728. Positive controls recover well (Fig. 2): chirp mass $R^2 = 0.920$–0.960, merger time $R^2 = 0.906$–0.928, SNR $R^2 = 0.741$–0.789, sky position $4.4°$–$10.3°$ mean angular error.

The machinery this study depends on is verifiably healthy; what it produces on the question of interest is presented next.

VI. Results: A Defended Null, Under the Corrected Formula

A. Headline result

At epoch 79 (last-10-epoch mean), POC_B's combination-space circular loss reads $\text{combo}_\text{A} = 0.9999$, $\text{combo}_\text{B} = 0.9895$ — both essentially at the ceiling value of 1 expected under random guessing. Table II reports angular recovery for all four down-selected models at $\lambda = 0.01$.

POC_A and POC_B are, by a systematic COLLAPSE grade ($\text{circ}_r > 0.9$ *and* $\text{MAE}_\text{ang} > 0.5$), genuinely mode-collapsed: POC_B's ψ predictions place 99.9% of 5,000 validation samples in one $10°$-wide histogram bin. TCN and $\text{CNN}_\text{attention}$ fail differently — noisy and uninformative rather than collapsed — but MAE_ang sits at the null for both. A direct forward-pass check makes the collapse concrete: three combo_A predictions from very different true angles ($-1.45, -2.22, +1.76$ rad) come back at $1.468, 1.496, 1.499$ rad regardless of input.

TABLE II
Periodic-head recovery, four down-selected models, $\lambda = 0.01$.

Head	Model	circ_r	MAE_ang	Grade
φ_c	POC_A	0.542	1.586	XX
φ_c	POC_B	0.975	1.535	**COLLAPSE**
φ_c	TCN	0.840	1.590	XX
φ_c	$\text{CNN}_\text{attention}$	0.386	1.616	XX
ψ	POC_A	0.972	0.801	**COLLAPSE**
ψ	POC_B	1.000	0.801	**COLLAPSE**
ψ	TCN	0.639	0.799	$\sim$
ψ	$\text{CNN}_\text{attention}$	0.396	0.785	$\sim$

B. Confound elimination

Loss wiring is confirmed correct in both training modes by direct code trace, not by inspecting a misleadingly generic loss-dictionary listing. The gradient path is confirmed live end to end (§V-A). The collapse is confirmed genuine, not a metric artifact, by direct histogram inspection (§VI-A). This study's own pre-registered interpretability gate (§VII) never clears at any λ tried, so these confound checks are reported directly against the trained checkpoints rather than inferred from a passing gate — an evidentiary substitution argued explicitly in §VIII-A.

C. Statistical significance

φ_c and ψ: 0 of 8 tests significant, most worse than chance. ι reads significant in all four models at the uncorrected $\alpha = 0.05$ level, but applying a 12-test Bonferroni correction ($0.05/12 \approx 0.00417$) leaves only $\text{CNN}_\text{attention}$'s $p = 0.0003$ surviving; POC_A($p = 0.0223$), POC_B($p = 0.0052$), and TCN($p = 0.0059$) do not. Inclination therefore

TABLE III
LABEL-PERMUTATION BOOTSTRAP ($N = 10{,}000$, ONE-SIDED). $z > 0$: BETTER THAN SHUFFLED-NULL.

Model	φ_c: z, p	ψ: z, p	ι: z, p
POC_A	-0.38, .649	-0.82, .793	$+2.00$, .0223
POC_B	-0.39, .649	-0.63, .740	$+2.58$, .0052
TCN	-1.18, .879	-0.50, .692	$+2.58$, .0059
$\text{CNN}_\text{attention}$	-1.97, .976	$+1.33$, .091	$+3.58$, .0003

reads statistically distinguishable from the shuffled null in one model under correction, and distinguishable only uncorrected in the other three — but §VI-G shows this reflects partial mode collapse, not a recoverable per-sample signal, in every case with an effect size well below this study's own materiality floor of 0.10 rad.

D. SNR stratification

No head/model combination clears the 0.10 rad floor in the high-SNR tercile for φ_c or ψ. TCN/φ_c is the sole monotonic-with-SNR improver ($\Delta = +0.0209$ rad at high SNR), but it is the one model whose STD_ratio never stabilizes at any λ (§VII), so its trend is dismissed as doubly uninterpretable, not merely small.

E. Loss trajectory and failure signature

Across all 80 epochs, combination-space circular loss is flat within noise for every model. The failure signature these checkpoints show is STD_ratio settling low from early training (§VII), not validation loss creeping upward late.

F. The perturbation-trace probe: parked, not closed

This study's calibration stage for the multi-step perturbation trace failed on its own pre-declared terms for every one of the four models: the mchirp positive control never reads directional even under favorable fresh-init conditions, so the probe's verdicts on the heads under test cannot be trusted. No salvage was attempted; the probe is stated as parked, not closed, and carried forward as future work (§VIII-F).

TABLE IV
SKY-POSITION ANGULAR SEPARATION.

Model	Bootstrap Δ (deg), z	High-SNR Δ	Verdict
POC_A	3.43, $+6.54\sigma$	$+3.01$	sub-floor
POC_B	3.03, $+5.51\sigma$	$+2.90$	sub-floor
TCN	2.86, $+5.21\sigma$	$+2.53$	sub-floor
$\text{CNN}_\text{attention}$	**12.58**, $+\mathbf{23.23}\sigma$	**+15.53**	**Recovery**

G. Inclination: partial mode collapse, not a small real signal or a clean noise floor

Inclination was used throughout as a same-model noise-floor calibration for judging φ_c/ψ's own band-to-band deviations, on the premise that its apparent failure was clean noise. That premise does not survive this study's bootstrap evidence (§VI-C): ι is not uninformative in the sense of pure random guessing. But neither is it a small, real, per-sample-recoverable signal, the reading originally given here. Instead, predictions cluster into a handful of preferred angles largely independent of the true value — partial mode collapse, quantified by this head's own circular resultant ($\text{circ}_r = 0.32$–0.60 across the four models), well short of the $\text{circ}_r > 0.9$ full-collapse grade used elsewhere for φ_c/ψ, but far above the ≈ 0 that either clean noise or genuine recovery would give. Tellingly, the ranking tracks each model's own φ_c/ψ collapse severity: POC_A/POC_B(COLLAPSE-graded on φ_c/ψ) show the highest inclination circ_r, TCN/$\text{CNN}_\text{attention}$(not collapsed there) the lowest — consistent with a shared training-dynamics origin across a model's periodic heads, not head-specific recoverable physics. No φ_c/ψ deviation, face-on to edge-on, clears the 0.10 rad floor in either direction for any model; a companion check on a non-angular control (chirp mass, banded identically) shows small band-to-band spread, ruling out the banding scheme itself as an alternative explanation.

H. A new finding: $\text{CNN}_\text{attention}$'s partial sky-localization skill

Sky position was carried throughout as a positive control; here it is tested for significance directly.

Three models show the same small-but-real, sub-floor pattern as inclination. $\text{CNN}_\text{attention}$clears the

materiality floor outright, and — since this model is independently known to memorize φ_c/ψ training-set structure without generalizing (§VIII-B) — the SNR discriminator resolves in favor of genuine recovery, not memorization: a hypothetical population-mean predictor scores only $\approx 89.4°$ (essentially at the null), and the actual error shrinks monotonically with SNR ($80.15° \to 77.95° \to 74.47°$). The leading mechanism hypothesis is cross-detector H1/L1 amplitude-ratio information via the sky-dependent antenna pattern — an envelope cue, not carrier phase — so this finding is peripheral to, and does not affect, the φ_c/ψ null.

VII. The Pre-Registered λ Retune

A. Why pre-register

This study adopted a strict pre-registration discipline from the outset, after a small cautionary episode of its own: on one same-day instance, an uncertainty-weight trajectory was misread by eye and had to be corrected with a mechanical check before any retune began. The criterion was written down — a Step 0 $\text{STD}_{\text{ratio}}$ gate, a Bonferroni-corrected bootstrap on the combination angles, a 0.10 rad effect-size floor, and an SNR-monotonicity check — before any $\lambda = 0.05/0.10$ result existed. Because the combination construction depends on both source vectors jointly, the gate requires all four readings (both heads, poc-mode and baseline-mode) to pass; a single failing reading keeps the whole round uninterpretable.

B. Outcome across three λ values

TABLE V
Step 0 gate, all three λ rounds, all four readings.

λ	Reading	final $\text{STD}_{\text{ratio}}$	Gate
0.01	$\text{POC}_\text{B}/\ \varphi_c$	0.583	FAIL
0.01	$\text{POC}_\text{B}/\ \psi$	0.251	FAIL
0.01	$\text{POC}_\text{A}/\ \varphi_c$	0.407	FAIL
0.01	$\text{POC}_\text{A}/\ \psi$	0.374	FAIL
0.05	$\text{POC}_\text{B}/\ \varphi_c$	0.315	FAIL
0.05	$\text{POC}_\text{B}/\ \psi$	0.531	FAIL
0.05	$\text{POC}_\text{A}/\ \varphi_c$	0.522	FAIL
0.05	$\text{POC}_\text{A}/\ \psi$	0.572	FAIL
0.10	$\text{POC}_\text{B}/\ \varphi_c$	0.930	PASS*
0.10	$\text{POC}_\text{B}/\ \psi$	0.421	FAIL
0.10	$\text{POC}_\text{A}/\ \varphi_c$	0.546	FAIL
0.10	$\text{POC}_\text{A}/\ \psi$	0.559	FAIL

*The one nominal PASS does not survive inspection. POC_B's φ_c at $\lambda = 0.10$ clears every numeric gate threshold, but its own training-time scatter plot shows two completely flat horizontal bands at the same angle modulo 2π — a literal two-point mode collapse present from epoch 5 through epoch 80, not real per-sample learning. $\text{STD}_{\text{ratio}}$ certifies raw-vector norm health, not angular diversity, and the two are not interchangeable: nothing about clearing the numeric threshold guarantees the scatter plot will look healthy.

With every reading at $\lambda = 0.10$ either failing outright or failing on inspection, the round closes: λ alone is insufficient to stabilize this pipeline's $\|v\|$-space for these heads. Steps beyond the gate never ran, because the gate never cleared at any λ, for any of the twelve readings. Per the pre-registered decision table, this is filed as **uninterpretable**, not as a null result and not as counter-evidence, on its own narrow terms — a negative result about the retune mechanism, not the basis for this paper's headline conclusion (§VI); §VIII-A reconciles the two.

VIII. Discussion

A. Reconciling two lines of evidence that do not collapse into one

This study's own two lines of evidence do not collapse into one, and the tension between them is argued here rather than asserted away. A pre-registered metric-health gate (§VII) was meant to license a clean, certified test of the degeneracy: no model, at any λ, ever passed it — there is no certified subset of any size to build that argument on.

The null in §VI instead rests on four independent methods — collapse/histogram inspection, a

gradient-health trace, a formal bootstrap, and population stratification — run uniformly across all four down-selected models, without excluding any on interpretability grounds. This substitution answers a different, and in some ways more direct, question than the gate does: not whether a specific combination vector's raw magnitude is well behaved, but whether these checkpoints' actual predictions carry recoverable signal — and the collapse and gradient-chain evidence rule out an implementation bug as the cause by a more direct method than gate-passage ever provided. It is also, we state plainly, a novel evidentiary structure that has not previously been scrutinized in this form. A skeptical reader is entitled to ask whether an ungated battery, assembled after the pre-registered path proved terminal, is a second attempt dressed as independent confirmation. It is a different, not a repeated, attempt: several of its components (the gradient-chain trace, the raw histogram inspection) are not calibrated against any threshold chosen after seeing data and simply report what is there — but the reactive order of construction is a real, stated limitation on how much independent confirmatory weight it should carry relative to a fully pre-registered test.

B. *Sufficiency of the architecture pool*

The five trunk families were chosen as five distinct inductive-bias hypotheses, not five samples from one family, and capacity is not the binding constraint: the memorization-gap check, run uniformly across all four down-selected models, shows $\mathrm{CNN}_{\mathrm{attention}}$'s training-split circular loss falling to 0.54–0.56 by epoch 80 while validation stays flat at ≈ 1.0 (train–val gap $+0.44$ to $+0.47$), against gaps an order of magnitude smaller for the other three models ($+0.003$ to $+0.05$). $\mathrm{CNN}_{\mathrm{attention}}$demonstrably has, and uses, spare capacity to fit sample-specific noise on φ_c/ψ and generalizes none of it — while simultaneously showing genuine, generalizing skill on sky position (§VI-H), illustrating that this model's capacity is head-specific, not uniformly wasted or informative.

C. *Relation to full-posterior methods*

This null does not reopen a settled question. Neural posterior estimation — conditional variational autoencoders [3], direct neural posterior-density approximations [23, 24], and conditional normalizing flows trained by simulation-based inference [4, 25–27] — does not attempt a single value for a parameter the likelihood only partially constrains; it learns an amortized approximation to the full posterior and reproduces multimodal, disconnected structure directly, exactly as a stochastic sampler does [12, 13]. For a genuinely degenerate target such as φ_c/ψ at low inclination, a posterior-estimating network faces no obstacle in principle. What this paper tests is narrower: whether a point-estimating regression head — still found in fast detection-and-characterization pipelines not designed as full posterior samplers [2] — fails safely or silently on a target of this kind. It fails silently, for a long stretch of this study's own diagnostic effort (§§V–VII), and in a second, more subtle way: a pre-registered interpretability gate can itself fail silently, passing a checkpoint that scatter-plot inspection reveals to be collapsed. The contribution is a documented account of both failure modes, and the diagnostic discipline a practitioner can reuse against either.

D. *Threats to validity*

In descending order of concern: no model here cleared the pre-registered gate at any λ, so the broader battery's trustworthiness rests on the argument of §VIII-A rather than gate certification; $\mathrm{POC}_{\mathrm{B}}$'s individual-head bootstrap numbers rest on ground-truth-assisted branch reconstruction, disclosed here explicitly; the inclination noise-floor comparison mixes two non-identically-distributed loss paths, a reason to read it descriptively; the perturbation-trace probe is open, not closed, with no salvage attempted; and the waveform-provenance and inclination-prior gaps remain unresolved (§VIII-F).

E. *Methodological lessons*

Aggregate metrics are not evidence until their mechanism is verified; controls must be validated

at the code level; pre-registration belongs inside engineering loops — confirmed here, sharply enough to add a fourth: **a passing interpretability gate is not proof that its subject matter is what the gate measures.** The $\lambda = 0.10$ false positive (§VII) demonstrated this directly: a checkpoint that cleared every numeric gate threshold was, on scatter-plot inspection, a literal two-point mode collapse. $\mathrm{STD}_{\mathrm{ratio}}$ measures raw-vector norm health, not angular diversity; the corrective, as with every lesson in this arc, is mechanism inspection — a scatter plot, not a summary statistic.

F. Future work

Inclination conditioning remains the natural next step given §III's analytic structure, strengthened rather than undermined by ι's newly-confirmed small real signal. Four items remain open, explicitly: a synthetic-data ablation disentangling $\mathrm{POC_B}$'s curriculum design from the physical degeneracy, a high-SNR (25–30) validation set or Fisher-matrix bound to separate degeneracy from a sensitivity floor, dataset regeneration from a version-controlled generator, and a 2D joint-scatter "ridge test" of predicted $(\hat{\varphi}_c, \hat{\psi})$ pairs, none of which this study ran. Two further items: confirming $\mathrm{CNN_{attention}}$'s sky-position mechanism directly, and recording a full software/environment manifest for this training campaign to support future reproducibility checks.

IX. Conclusion

We set out to determine whether the coalescence phase and polarization angle of compact-binary signals are recoverable from strain by direct neural regression, individually or in their physically motivated sum/difference combinations. The answer is **no**. Every configuration tested converged to the optimal constant or noise-like predictor at the analytic random-guessing error on both angles, while the same shared representation recovered chirp mass, merger time, signal-to-noise ratio, and — for one architecture — a genuine partial sky-localization skill, from the same features. No model here passed the pre-registered metric-health gate; the null instead rests on convergence across direct collapse inspection, a gradient-health trace, a formal permutation bootstrap, and population stratification, applied uniformly across every down-selected model rather than to a certified subset — a defensible basis for the conclusion, argued explicitly in §VIII-A, but one that has not yet been through independent adversarial scrutiny. Two findings exist only because this study looked further than strictly necessary: $\mathrm{CNN_{attention}}$'s real, SNR-verified sky-localization skill, and inclination's small-but-real departure from the clean-noise-floor role this study's own methodology initially assumed for it. Neither changes the central verdict; both sharpen, honestly, what it does and does not claim. The degeneracy remains, on the evidence presented here, effectively exact for this population as a point-estimation problem.

Appendix: Claim-to-artifact map

This appendix follows the traceability convention of the accompanying code repository: every quantitative claim above is backed by a versioned artifact, listed here with its repository-relative path. All paths relative to `experiments/phic_psi_poc-redo/`.

Claim / table / figure	Artifact
Analytic sweep (§3, Fig. 1)	`sweep_1-1_ratio_vs_corr.[png,pdf]`
Gradient-health trace (§5.1.3)	`diagnostic_output/diagnostic_checks_20260916_113913.log`
Gradient-chain numbers (§5.2)	`diagnostic_output/diagnostic_report_20260916_113847.md`
Table II, Bonferroni correction	`bootstrap_output/bootstrap_ang_mae_20260916_113955.md`
SNR stratification (§6.4)	`snr_output/snr_stratification_20260916_112730.md`
Table IV, inclination scatter	`inclination_output/inclination_scatter_20260923_132229.[png,pdf]` **partial-collapse scatter**; `inclination_output/bootstrap_snr_separation_20260923_132229.md`
Table IV, sky-position finding (§6.6)	`bootstrap_output/bootstrap_sky_separation_20260916_152359.md`, `snr_output/snr_stratification_sky_position_20260916_153412.md`
Table V, $\mathrm{STD_{ratio}}$ false-positive	`diagnostic_output/diagnostic_logvar_gate_20260916_245602.md`; **scatter PNGs**
Memorization-gap numbers (§8.2)	`diagnostic_output/redo_models_train_val_loss.[png,pdf]`
Undisclosed POC$_{\mathrm{B}}$ reconstruction property	`bootstrap_ang_mae.py`, formulas cross-referenced in §A.8

One number in this paper is derived rather than printed in its source artifact: the Bonferroni threshold 0.00417 in §VI-C is $0.05/12$, arithmetic on the twelve bootstrap tests, not a value stated in `bootstrap_ang_mae_20260916_113955.md` itself.

Data and code availability

All training/evaluation code, configuration files, and analysis scripts are versioned in this repository. The dataset file (`combined_repackaged.hdf`) is not itself version-controlled or regenerable from a tracked generator script; this gap is stated as an open item in §VIII-F, not closed by this paper. This paper has not yet been through independent adversarial review; the evidentiary substitution of §VIII-A is the claim most in need of that scrutiny.

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